Monitoring System Requirements

Software Requirements Specification

IEEE 830-1998 outline

DRAFT — human review required

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1 Introduction

1.1 Purpose

This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

This SRS describes the software's required behavior for users, administrators and the Project Analyst reviewing the specification.

1.2 Scope

Design a monitoring system that collects authentication events from laboratory servers. The system must detect repeated failed sign-in attempts. Security operators can review detection results. The system must not lock accounts automatically. Leave the detection threshold open for review.

1.3 Definitions, acronyms and abbreviations

No project-specific terms were supplied.

1.4 References

Reference	Source
Format	IEEE Std 830-1998, clauses 5 and Annex A (document organization).

1.5 Overview

Section 2 summarizes the product, stakeholders and open decisions. Section 3 contains the uniquely identified requirements. Detailed evidence, acceptance procedures are available in the Normal report. Appendices A-C contain derived threat hypotheses, risk assessment and the proposed SDLC workflow.

2 Overall description

2.1 Product perspective

Design a monitoring system that collects authentication events from laboratory servers.

2.2 Product functions

Functional topics are summarized below; section 3 retains the full requirements.

- collect authentication events from laboratory servers - FR-001
- detect repeated failed sign-in attempts - FR-002
- allow security operators to review detection results - FR-003
- not lock accounts automatically - FR-004

2.3 User characteristics and stakeholders

Labels group stakeholders for readability. They do not combine permissions.

Education, training and experience requirements remain unspecified.

Role	Source user groups	Responsibility
Operator	Security operators	Performs only the role's source-defined operations in section 3.
Project Analyst	Document review role	Reviews requirements and tracks open decisions; this role grants no product access.

2.4 Constraints

- The system must not lock accounts automatically.

2.5 Assumptions and dependencies

Open decisions requiring confirmation:

- Leave the detection threshold open for review.
- What conditions define the requested detections and their alert triggers?
- How are the stated roles and authorized permissions assigned?

3 Specific requirements

Each requirement is stated once with its original identifier and wording. Cross-references identify requirements covering more than one concern.

3.1 External interface requirements

3.1.1 User interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.2 Hardware interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.3 Software interfaces

Related requirements: FR-001. Interface details beyond those requirements remain unspecified.

3.1.4 Communications interfaces

No specific interface requirement was supplied; confirmation is required.

3.2 Functional requirements

FR-001 · Must · explicit

The system shall collect authentication events from laboratory servers.

Traceability: E-001, TEST-001

FR-002 · Must · explicit

The system shall detect repeated failed sign-in attempts.

Traceability: E-002, TEST-002

FR-003 · Must · explicit

The system shall allow security operators to review detection results.

Traceability: E-003, TEST-003

FR-004 · Must · explicit

The system shall not lock accounts automatically.

Traceability: E-004, TEST-004

3.3 Performance requirements

No separate requirement was specified in the supplied brief.

3.4 Logical database requirements

No separate requirement was specified in the supplied brief.

3.5 Design constraints

No separate requirement was specified in the supplied brief.

3.6 Software system attributes

3.6.1 Reliability

No separate requirement was specified in the supplied brief.

3.6.2 Availability

No separate requirement was specified in the supplied brief.

3.6.3 Security

Also covered by: FR-001, FR-003.

3.6.4 Maintainability

No separate requirement was specified in the supplied brief.

3.6.5 Portability

No separate requirement was specified in the supplied brief.

3.7 Other requirements

No separate requirement was specified in the supplied brief.

A Derived threat hypotheses

These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

These are planning hypotheses, not observed incidents or additional approved controls.

A.1 THR-001: Monitoring and security alerts

Scenario: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Relevant suspicious activity could remain undetected or receive delayed investigation.

Linked requirements: FR-001, FR-002, FR-003

Original evidence: E-001, E-002, E-003

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.2 THR-002: User identities and authenticated access

Scenario: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Unauthorized use of a legitimate identity could expose resources or alter system state.

Linked requirements: FR-001

Original evidence: E-001

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.3 THR-003: Integration trust paths

Scenario: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Untrusted integration data could cause incorrect decisions or hide security events.

Linked requirements: FR-001

Original evidence: E-001

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

B Risk assessment

Risk treatments are proposals for review. Unknown likelihood and impact ratings mean unassessed; no numeric confidence or severity is inferred.

- Decision Q-ea57354867fa is TBD.

B.1 RISK-001

Derived risk affecting Monitoring and security alerts: Relevant suspicious activity could remain undetected or receive delayed investigation.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Proposed treatment: Evaluate the stated detection duties with representative positive and negative events; confirm undecided thresholds, alert priorities and investigation responsibilities. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-001, FR-002, FR-003

Original evidence: E-001, E-002, E-003

B.2 RISK-002

Derived risk affecting User identities and authenticated access: Unauthorized use of a legitimate identity could expose resources or alter system state.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Proposed treatment: Review impersonation and compromised-identity paths against the stated access duties; confirm any additional identity or session protections with the stakeholder. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-001

Original evidence: E-001

B.3 RISK-003

Derived risk affecting Integration trust paths: Untrusted integration data could cause incorrect decisions or hide security events.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Proposed treatment: Review trust assumptions at the specified integration boundaries with missing or altered input fixtures; confirm the required failure handling with the owner. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-001

Original evidence: E-001

B.4 RISK-004

An unresolved source decision may delay validation: Leave the detection threshold open for review.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: The source leaves this decision unresolved; no delivery or technical conclusion is inferred.

Proposed treatment: Obtain a stakeholder decision before baselining affected behavior; retain the unknown and revalidate resulting changes.

Linked requirements:

Original evidence: E-005

C SDLC recommendation and delivery workflow

Proposed lifecycle: DevSecOps

Security decisions benefit from security review and tests throughout delivery (E-001); monitoring and alert behavior needs validation with operational feedback; connected systems need integration verification before broader rollout. Derived assurance factors support DevSecOps over Hybrid, Incremental, Spiral, Agile, V-Model, Waterfall due to integrated security and continuous operational feedback. Unknown factors include budget, delivery frequency, failure consequences, legacy system integration, regulation, schedule, stability, uncertainty, and verification methods; these are not assumed or resolved.

Recommendation requires human confirmation of delivery constraints and team capacity. Suitability points are deterministic rule scores, not probabilities.

Policy version: sequential-sdlc-2.0

Unconfirmed decision factors: budget, delivery, failure consequences, feedback, legacy, regulation, schedule, stability, uncertainty, verification

- Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.
- Proposed suitability recommendation; confirm delivery constraints and team capacity before adopting this lifecycle. Missing factors are not assumed.

C.1 Ranking and source evidence

- security_assurance = integrated security review (derived). Security decisions benefit from security review and tests throughout delivery. Original evidence: E-001.
- operational_validation = monitoring feedback (derived). Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. Original evidence: E-001, E-002, E-003, E-005.
- integration_assurance = integration validation (derived). Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. Original evidence: E-001.

Candidate	Suitability points
DevSecOps	60
Hybrid	57
Incremental	54
Spiral	51

Candidate	Suitability points
Agile	40
V-Model	40
Waterfall	40

C.1.1 DevSecOps suitability reasons

- Derived security_assurance (+12 points): Security decisions benefit from security review and tests throughout delivery. (E-001)
- Derived operational_validation (+8 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-001, E-002, E-003, E-005)

C.1.2 Hybrid suitability reasons

- Derived security_assurance (+8 points): Security decisions benefit from security review and tests throughout delivery. (E-001)
- Derived operational_validation (+5 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-001, E-002, E-003, E-005)
- Derived integration_assurance (+4 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-001)

C.1.3 Incremental suitability reasons

- Derived operational_validation (+6 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-001, E-002, E-003, E-005)
- Derived integration_assurance (+8 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-001)

C.1.4 Spiral suitability reasons

- Derived security_assurance (+5 points): Security decisions benefit from security review and tests throughout delivery. (E-001)
- Derived integration_assurance (+6 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-001)

C.1.5 Agile suitability reasons

- No differentiating input evidence

C.1.6 V-Model suitability reasons

- No differentiating input evidence

C.1.7 Waterfall suitability reasons

- No differentiating input evidence

C.2 Methodology phases

Activities reference the canonical requirements and proposed verification work. Human gates are pending decisions, not approval records.

C.2.1 Requirements and verification planning

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003, FR-004

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review and baseline the source-defined work group: FR-001: The system shall collect authentication events from laboratory servers.; FR-002: The system shall detect repeated failed sign-in attempts.; FR-003: The system shall allow security operators to review detection results.
- Activity: Review and baseline the source-defined work group: FR-004: The system shall not lock accounts automatically.
- Activity: Record unresolved source decision: Leave the detection threshold open for review.
- Activity: Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.
- Deliverable: Versioned requirements baseline
- Deliverable: Unit, integration, system and acceptance test plan
- Deliverable: Unresolved-decision record
- Entry criterion: Authoritative inputs and canonical requirements are available for review.
- Exit criterion: A human reviewer has recorded the baseline decision and disposition of unresolved scope.
- Security check: Review design and test evidence for violations of source-defined permissions and exclusions.

C.2.2 Design and test design

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003, FR-004

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Design the implementation for the source-defined work group: FR-001: The system shall collect authentication events from laboratory servers.; FR-002: The system shall detect repeated failed sign-in attempts.; FR-003: The system shall allow security operators to review detection results.
- Activity: Design the implementation for the source-defined work group: FR-004: The system shall not lock accounts automatically.
- Activity: Refine the early unit, integration, system and acceptance test plan without choosing unresolved product technologies.
- Deliverable: Proposed design linked to canonical requirements
- Deliverable: Unit, integration, system and acceptance test plan
- Entry criterion: The requirements baseline and early test plan have been reviewed.
- Exit criterion: A human reviewer has recorded the design decision and test-plan readiness.

- Security check: Review design and test evidence for violations of source-defined permissions and exclusions.

C.2.3 DevSecOps delivery cycle

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003, FR-004

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review and prioritize the backlog for the next iteration without changing source-defined scope.
- Activity: Implement the source-defined work group: FR-001: The system shall collect authentication events from laboratory servers.; FR-002: The system shall detect repeated failed sign-in attempts.; FR-003: The system shall allow security operators to review detection results.
- Activity: Implement the source-defined work group: FR-004: The system shall not lock accounts automatically.
- Activity: In an authorized isolated test environment, execute TEST-001: Given: A laboratory server generates an authentication event When: The monitoring system is active and configured to receive events Expected: The system shall record the authentication event in its data stream
- Activity: In an authorized isolated test environment, execute TEST-002: Given: A user attempts to sign in multiple times with incorrect credentials When: The monitoring system processes authentication logs sequentially Expected: The system shall identify a sequence of failed sign-ins exceeding the threshold for repetition
- Activity: In an authorized isolated test environment, execute TEST-003: Given: An operator is authenticated and authorized to access the monitoring interface When: The operator requests to view detection results Expected: The requested detection results are returned in a visible format
- Activity: In an authorized isolated test environment, execute TEST-004: Given: A user account experiences multiple failed sign-in attempts When: The monitoring system detects the pattern but is configured to prevent automatic account locking Expected: No automated account lockout action shall occur
- Activity: Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
- Activity: Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.
- Activity: Review security-check evidence for every delivery change and record a human release decision; keep unresolved tooling and operational choices open.
- Deliverable: Requirement-linked implementation increment
- Deliverable: Recorded test evidence
- Deliverable: Reviewed feedback and next-iteration backlog
- Entry criterion: Prior work, source scope and the reviewed test plan are available for iteration planning.
- Exit criterion: A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.
- Security check: Review design and test evidence for violations of source-defined permissions and exclusions.

C.2.4 Release and handoff

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003, FR-004

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review the exact requirements baseline, implementation links and verification evidence before a release decision.
- Activity: Carry unresolved source decisions into handoff; generated text is not human approval.
- Deliverable: Requirements and verification evidence package
- Deliverable: Pending human release decision
- Deliverable: Outstanding-decision record
- Entry criterion: Verification evidence and unresolved decisions are available for human review.
- Exit criterion: A human has recorded the release/handoff decision for the reviewed content.
- Security check: Review design and test evidence for violations of source-defined permissions and exclusions.

Detailed acceptance criteria, evidence, architecture and the SDLC plan are available in the supporting report and canonical JSON.

Canonical content SHA-256: 6cf4e36301bbc72313e211275b5f37145bb6d192f6fd4882cbcf002d14cde621